

KRESHCHENSKOYE, Russian Federation

WMO: 295240

Lat: 55.8593N

Lon: 80.0608E

Elev: 129

StdP: 99.79

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.2	-33.8	-39.9	0.1	-36.9	-36.6	0.1	-33.6	7.8	-10.2	6.9	-8.8	0.9	250	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	28.4	19.2	26.8	18.6	25.3	17.9	20.9	26.1	19.9	25.1	18.9	23.6	3.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.0	23.6	17.9	13.1	22.5	17.0	12.3	21.6	61.2	26.0	57.7	25.1	54.4	23.6	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.4	5.6	-41.6	31.3	3.7	2.5	-44.2	33.1	-46.4	34.6	-48.5	36.0	-51.1	37.9		
			-41.2	22.5	3.5	0.9	-43.8	23.2	-45.8	23.7	-47.8	24.3	-50.4	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-18.7	-15.6	-7.3	2.6	10.7	16.0	17.9	14.9	9.0	2.3	-7.9	-15.3
	DBStd	14.12	8.33	7.45	6.39	5.86	5.36	4.15	3.21	3.60	4.29	4.99	7.88	8.64
	HDD10.0	4078	891	717	537	229	60	6	0	4	68	243	537	786
	HDD18.3	6468	1149	951	795	471	242	89	47	116	280	498	787	1044
	CDD10.0	721	0	0	0	8	81	187	245	157	39	4	0	0
	CDD18.3	69	0	0	0	0	4	20	34	10	1	0	0	0
	CDH23.3	829	0	0	0	1	96	266	332	119	15	0	0	0
	CDH26.7	156	0	0	0	0	19	53	66	17	1	0	0	0
(14) Wind	WSAvg	2.6	2.5	2.6	3.0	3.1	3.0	2.2	2.0	2.0	2.4	2.7	2.9	2.8
(15) Precipitation	PrecAvg	436	23	17	18	22	34	55	65	60	36	42	35	30
	PrecMax	581	44	28	47	66	86	114	160	118	93	92	68	61
	PrecMin	288	3	1	3	4	7	3	6	13	7	18	8	8
	PrecStd	76	11	6	11	14	18	32	34	23	20	21	13	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	1.1	7.0	21.6	29.0	30.0	30.7	28.8	25.5	19.2	7.1	1.2
		MCWB	-0.7	-0.1	3.1	12.5	16.5	19.8	21.3	20.2	16.7	10.9	5.1	0.2
	2%	DB	-2.3	-0.9	4.5	17.9	25.9	28.3	28.4	26.5	22.1	15.4	4.9	0.0
		MCWB	-3.1	-1.8	1.4	10.8	15.3	18.7	20.1	19.2	14.7	9.4	3.4	-0.8
	5%	DB	-4.4	-2.8	2.8	14.6	23.5	26.4	26.8	24.5	19.5	12.2	3.1	-1.8
		MCWB	-5.0	-3.6	0.8	8.9	14.2	18.3	19.4	18.0	13.4	7.9	2.0	-2.4
	10%	DB	-6.5	-4.6	1.4	11.6	20.7	24.5	25.1	22.5	17.1	9.6	1.1	-3.8
		MCWB	-7.1	-5.5	-0.3	6.9	12.9	17.3	18.8	16.9	12.0	6.6	0.2	-4.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	0.1	3.5	13.2	18.0	21.5	22.5	21.4	17.4	11.8	5.8	0.5
		MCDB	0.0	0.9	6.3	19.6	26.5	27.2	28.4	26.5	22.9	17.2	6.6	1.0
	2%	WB	-3.1	-1.7	2.0	11.3	16.2	20.1	21.3	20.0	15.4	10.1	3.6	-0.7
		MCDB	-2.4	-0.9	3.9	17.2	24.0	26.2	26.5	25.3	20.9	14.6	4.7	-0.1
	5%	WB	-5.1	-3.5	0.9	9.2	14.7	19.0	20.4	18.7	13.9	8.3	2.0	-2.5
		MCDB	-4.5	-2.9	2.6	14.3	22.1	24.7	25.3	23.1	18.6	11.7	3.1	-1.8
	10%	WB	-7.1	-5.5	-0.2	7.2	13.3	18.0	19.4	17.5	12.5	6.7	0.4	-4.5
		MCDB	-6.6	-4.7	1.4	11.2	19.5	23.5	23.9	21.6	16.3	9.5	1.1	-3.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.7	10.1	9.8	11.7	11.7	10.9	11.2	10.5	7.9	6.8	8.1
		MCDBR	8.7	7.4	9.4	14.4	16.5	15.1	13.9	14.6	15.3	13.5	6.2	7.3
		MCWBR	8.6	7.0	7.4	8.6	7.7	7.2	6.7	8.0	8.6	8.2	5.3	7.1
	5% WB	MCDBR	8.9	7.0	8.1	13.9	15.0	13.6	12.2	12.9	13.6	12.0	6.2	7.1
		MCWBR	8.9	6.8	6.6	8.6	7.7	7.0	6.5	7.7	8.0	7.7	5.5	7.1
(40) Clear-Sky Solar Irradiance	taub		0.286	0.318	0.371	0.408	0.404	0.405	0.438	0.411	0.366	0.349	0.315	0.287
	taud		2.018	1.989	1.919	2.071	2.225	2.295	2.197	2.255	2.334	2.237	2.071	1.962
	Ebn at Noon		579	710	760	801	835	839	798	796	784	683	534	458
	Edh at Noon		67	102	141	141	129	122	132	117	94	80	63	55
(44) All-Sky Solar Radiation	RadAvg		0.87	1.89	3.39	4.60	5.33	5.89	5.31	4.31	2.95	1.48	0.71	0.54
	RadStd		0.07	0.11	0.18	0.36	0.51	0.58	0.38	0.37	0.32	0.15	0.13	0.07

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page